

Moshe Mash, Ph.D.

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Research Interests

My research centers on environments shared by humans and computer agents: how they interact, coordinate, and reach agreements, and how to design agents that act well alongside people. I combine game theory and mechanism design with machine learning to model, predict, and shape behavior in such settings, from joint exploration and negotiation to coalition formation and fair division. I am developing a new direction that extends this to LLM-based agents: learning what an agent should communicate to a human, and how, from human preference, in mixed human-agent settings such as autonomous and human-driven vehicles.

A related line of work, begun during my postdoc at Carnegie Mellon, studies the people who build AI systems: capturing and modeling the workflows of data scientists as they develop machine learning models, and predicting the moments where they get stuck so that tools can offer targeted support.

A third, applied strand of my research uses large language models and modern NLP for text analysis in specialized domains: authorship attribution, automatic citation generation, and semantic cross-referencing of Hebrew and rabbinic texts (with DICTA), and computational analysis of legal and patent text, including textual studies of disparate outcomes in the patent system. Both involve fine-tuning LLMs for prediction, clustering, and semantic analysis of domain-specific corpora.

Education

Ph.D., Software and Information Systems Engineering | Ben-Gurion University of the Negev | 2014–2018

- **Dissertation:** Reaching Fair Agreements in Group Settings
- **Advisor:** Prof. Ya'akov (Kobi) Gal

Dissertation research on how people and computer agents reach fair agreements in group settings. One line of work developed algorithms for fair division: a general framework for computing envy-free rent divisions that optimize a criterion of social justice, identifying the maximin solution as the fairest. It was validated in a user study with real users of the Spliddit website and has been deployed on Spliddit since 2015 (ACM EC 2016 Best Paper Award; JACM;

Communications of the ACM Research Highlight). A second line studied coalition formation in weighted voting games, combining cooperative game theory with machine learning to predict how people form coalitions, and designing an agent that negotiates with people and outperformed human players (IJCAI 2017; ACM TIST). Mentored undergraduate students and advised final-year projects throughout the Ph.D.

M.Sc., Computer Science (AI specialization) | Bar-Ilan University | 2011-2013

- **Thesis:** Joint Exploration with Self-Interested Agents
- **Advisor:** Prof. David Sarne

Thesis research on joint exploration by self-interested agents: settings in which several agents search for opportunities whose values are uncertain and costly to reveal, and share what they find. Using equilibrium analysis, the work showed that mechanisms intended to enhance cooperation, such as longer search horizons or more cooperating agents, can degrade individual and overall utility when agents are self-interested, with implications for the design of multi-agent markets and systems (Artificial Intelligence Journal; IAT 2012).

B.Sc., Software Engineering | Shenkar College of Engineering and Design | 2007-2011

Awards & Recognition

- **Best Paper Award** – ACM Conference on Economics and Computation (2016)
 - **Communications of the ACM Research Highlight** – Which Is the Fairest (Rent Division) of Them All? (2018)
 - **Excellence in Teaching Award** – Ben-Gurion University (2016, 2017)
 - **Ph.D. Scholarship for Academic Excellence** – Ben-Gurion University (2014)
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Academic Positions

Postdoctoral Researcher | Carnegie Mellon University, Robotics Institute | 2019-2022

Hosted by Prof. Reid Simmons and Dr. Stephanie Rosenthal. My research studied the people who build AI systems: how machine learning practitioners and data scientists work as they develop models, and where they get stuck. I developed DSWorkflow, a framework that captures data scientists' workflows from Jupyter notebooks, reconstructs the order in which code was executed, and combines it with qualitative data, and used it in studies of data scientists building machine learning models (CHI). Building on that data, I developed a model that predicts

stuckness from real-time indicators such as code artifacts, and an algorithm that times an intervention as close as possible to the onset of stuckness (VL/HCC).

I mentored students in AI and human-computer interaction research. The research was funded by J.P. Morgan, which renewed the award for a second consecutive year based on the project's results.

Professional Experience

Principal AI Researcher & Founder | MashInnovateAI | 2024-present

- Independent AI research & development consultancy: helps organizations identify where AI can solve their hardest problems, then designs and delivers the solution end to end - applied ML, NLP, LLM fine-tuning, RAG, and agentic and multi-agent systems - from proof of concept to production deployment.
- Clients include technology companies looking to integrate AI into their products and workflows, startups, and universities and academic research groups.

AI Researcher | DICTA - The Israel Center for Text Analysis | 2022-present

- Lead the center's NLP and LLM research on large collections of Hebrew and rabbinic texts: set the research agenda, design the methods, and take systems from prototype to tools in active use by scholars.
- Authorship attribution: built models that predict the author of texts with historically disputed attribution from textual signals alone, reaching high accuracy and resolving long-debated questions of authorship.
- Citation detection and cross-referencing: designed systems that automatically detect, extract, and link citations of classical sources across a large corpus, connecting documents that quote or paraphrase one another.
- Document understanding with LLMs: fine-tuned LLMs and trained custom text embeddings with contrastive learning for domain-specific corpora, powering information extraction, semantic search, text similarity, clustering, and question answering over long, complex documents.
- Linguistic analysis: built deep-learning models for morphological, syntactic, and semantic analysis of long, complex documents at scale.

- Infrastructure and deployment: built the end-to-end training and inference pipeline, with tooling to analyze model outcomes and inspect the training process, and deployed the resulting document-understanding systems to production.

AI Researcher | Diagnostic Robotics | 2019

- Developed machine learning models for the healthcare domain, with applications in risk prediction (e.g., ER diagnoses, hospitalization forecasting).
- Collaborated with clinicians and researchers to refine AI-driven predictive healthcare decision-support systems.

AI Researcher | Amdocs | 2018-2019

- Led the development of a churn-prediction system for at-risk maintenance contract renewals, saving millions of dollars.
- Designed, built, and deployed the system from scratch, including problem definition, algorithm design, evaluation, and deployment pipeline.

Research Assistant | Weizmann Institute of Science | 2011-2012

- Developed image-processing algorithms for 3D biomedical images.
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Teaching & Advising

Ben-Gurion University of the Negev – Software and Information Systems Engineering Dept.

- Teaching Assistant, Databases – Fall-Spring 2016, 2017, 2018
- Teaching Assistant, Analysis and Design of Software Systems – Spring-Summer 2016, 2017, 2018
- Teaching Assistant, Intelligent Systems – Spring-Summer 2015
- Teaching Assistant, Decision Support Systems – Fall-Spring 2015
- Departmental Seminar Coordinator – 2015, 2016, 2017, 2018
- Final Projects Coordinator (4th Year) – 2015, 2016, 2017, 2018
- Final Projects Adviser (Undergraduate) – 2015, 2016

Tel-Aviv Yafo Academic College – Information Systems Dept.

- Lecturer, Databases – Spring-Summer 2015, Fall-Spring 2015

- Lecturer, Game Theory – Spring–Summer 2015
- Teaching Assistant, Databases – Spring–Summer 2014

Bar-Ilan University

- Teaching Assistant, Statistical Methods in Computer Science – Spring–Summer 2014
- Teaching Assistant, Object-Oriented Programming – Fall–Spring 2013, 2014
- Teaching Assistant, Machine Learning – Fall–Spring 2014
- Teaching Assistant, Introduction to Computing – Fall–Spring 2013

Shenkar College of Engineering and Design

- Teaching Assistant, Introduction to Programming Systems – Spring–Summer 2012
- Teaching Assistant, Introduction to Computer Science – Fall–Spring 2012
- Teaching Assistant, Introduction to Object-Oriented Programming – Fall–Spring 2012

College of Management (Academic)

- Teaching Assistant, Object-Oriented Programming – Spring–Summer 2012

High School Teaching (Israel Ministry of Education)

- Mathematics Teacher (2010–2013) – taught Bagrut levels 3–5 points, specializing in students with learning disabilities (via Nitzan Association).

Professional Service

- **Reviewer:** AAAI, IJCAI, AAMAS, KDD, CHI, EC, ECAI, Group Decision & Negotiation
 - Developed voting algorithm for Best Paper selection (IAAI 2018)
 - Organized departmental seminars at Ben-Gurion University
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Publications

Journal Articles

- Gerhardt D., Marcowitz-Bitton M., Schuster W.M., Elmalech A., Suissa O., Mash M. (2026). Gendered Words and Patent Grant Rates: A Textual Analysis. *Northwestern Journal of Technology and Intellectual Property*, 23(3), Article 2.
- Raphael I, Xiong Z, Sneiderman CT, Raphael RA, Mash M., et al. The T cell receptor landscape of childhood brain tumors. *Science Translational Medicine*. 2025 Mar 19; 17(790): eadp0675.
- Mash, M., Fairstein, R., Bachrach, Y., Gal, K., & Zick, Y. (2020). Human-Computer Coalition Formation in Weighted Voting Games. *ACM Transactions on Intelligent Systems and Technology (TIST)*.
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2017). Which Is the Fairest (Rent Division) of Them All? *Journal of the ACM (JACM)*. *Selected as a Communications of the ACM Research Highlight, 2018*.
- Rochlin, I., Sarne, D., & Mash, M. (2014). Joint Search with Self-Interested Agents and the Failure of Cooperation Enhancers. *Artificial Intelligence Journal (AIJ)*.

Conference Proceedings

- Mash, M., Oryol, S., Rosenthal, S., & Simmons, R. (2022). Predicting Data Scientist Stuckness During the Development of Machine Learning Classifiers. *VL/HCC*.
- Mash, M., Rosenthal, S., & Simmons, R. (2021). DSWorkflow: A Framework for Capturing Data Scientists' Workflows. *CHI*.
- Mash, M., Bachrach, Y., Gal, K., & Zick, Y. (2017). How to Form Winning Coalitions in Mixed Human-Computer Settings. *IJCAI*.
- Gal, K., Mash, M., Procaccia, A., & Zick, Y. (2016). Which Is the Fairest (Rent Division) of Them All? *EC*. (Best Paper Award)
- Mash, M., Lin, R., & Sarne, D. (2014). Peer-Design Agents for Reliably Evaluating Distribution of Outcomes in Environments Involving People. *AAMAS*.
- Mash, M., Rochlin, I., & Sarne, D. (2012). Join Me with the Weakest Partner, Please. *IAT*.